PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

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Abstract

This pilot examines how inter-individual differences moderate the effectivity of cyber-manipulation on political opinions using an ontology-driven multi-agent simulation pipeline. Run 6 uses an attacked-only profile-panel design in which one fixed misinformation leaf is applied to each pseudoprofile across multiple political-opinion leaves. Attack effectivity is defined as within-profile post-minus-baseline movement, with absolute shift $A_{ik} = |\text{post}_{ik} - \text{baseline}_{ik}|$ as the primary outcome so that issue-specific signed movements do not cancel out. The pilot generated 50 pseudoprofiles, 200 attacked opinion rows, and 4 repeated attacked outcomes per profile. Mean absolute attacked shift was 41.65, mean signed shift was 22.05, mean attack realism was 0.69, and mean post-exposure plausibility was 0.75. The structural model is now a repeated-outcome path SEM in which multiple attacked opinion deltas are regressed on profile moderators simultaneously. To compute a profile-level susceptibility summary without inheriting multicollinearity from a small-sample OLS, a cross-validated ridge aggregation is fit after the run and converted into a post hoc empirical susceptibility index. The pilot should therefore be read as methodological validation and exploratory evidence rather than as claim-ready causal estimation.

1 Introduction

Political influence campaigns increasingly operate in the space between mass communication, personalized persuasion, and platform-native misinformation. Rather than relying on technical compromise alone, they often work through selective framing, pseudo-consensus, strategic ambiguity, distrust activation, and identity-congruent narrative design (Bennett & Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017; Pennycook & Rand, 2019, 2021; Vosoughi et al., 2018). This makes the key research problem not only whether content is false, but whether the cognitive conditions of judgement are being shaped in ways that weaken self-directed political belief formation.

That concern is closely related to recent discussions of cognitive warfare and cognitive superiority in security policy. NATO-oriented analyses increasingly frame the contested terrain as one in which human attention, interpretation, trust, and decision-making become operationally relevant dimensions of conflict and competition (Johns Hopkins University & Imperial College London, 2021; Hung & Hung, 2022; Paulauskas, 2024). The present study adopts *cognitive sovereignty* as the substantive framing for whether political opinion formation remains self-directed under digital adversarial pressure.

The research question is straightforward: *How do inter-individual differences moderate the effectivity of cyber-manipulation on political opinions?* The methodological challenge is less

straightforward. A real-world dataset that cleanly aligns profile structure, attack-vector semantics, repeated political-opinion items, and post-exposure response states is difficult to obtain, ethically constrained, and often poorly matched to ontology-level modeling. This makes a transparent, auditable simulation architecture useful as a methodological instrument, provided the architecture is explicit enough to be criticized and improved (Cioffi-Revilla, 2002; Rosseel, 2012).

2 Materials and Methods

2.1 Ontology-Constrained Pilot Design

Run 6 uses three hierarchical ontologies: PROFILE, ATTACK, and OPINION. The ontologies remain hierarchical in storage, but estimation is leaf-based. PROFILE combines continuous leaves (e.g., age and Big Five means) with categorical leaves (e.g., sex). ATTACK contributes one fixed pilot leaf, `ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing`. OPINION contributes repeated leaf nodes from the focused domain Defense_and_National_Security.

The design is attacked-only. Each pseudoprofile is crossed with multiple opinion leaves, so one fixed attack leaf is connected to multiple attacked deltas for the same individual profile. This architecture is more coherent with the moderation question than a treatment-versus-control setup when the research focus is heterogeneity of attack effectivity under exposure rather than incremental exposure effects relative to no exposure.

2.2 Baseline, Exposure, and Post-Exposure Assessment

For each profile i and opinion leaf k, the pipeline elicits a baseline opinion and a post-exposure opinion on a signed integer scale from -1000 to $+1000$. Exposure texts are generated as realistic social-media-native manipulative messages, then audited for realism. Post-exposure responses are audited for coherence and boundedness. This means every analyzed row is not merely a model output, but a generated output that has passed a second review layer designed to detect implausible reversals, coarse rounding, and issue-incoherent response shifts.

2.3 Effectivity Outcome and Profile-Level Susceptibility

The signed attacked shift for profile i on opinion leaf k is

$$\Delta_{ik} = \text{post}_{ik} - \text{baseline}_{ik}, \tag{1}$$

and the primary effectivity outcome is the absolute attacked shift

$$A_{ik} = |\Delta_{ik}|. \tag{2}$$

The profile-level observed effectivity summary is the mean attacked shift magnitude

$$E_i = \frac{1}{K} \sum_{k=1}^{K} A_{ik}, \tag{3}$$

where K is the number of repeated opinion leaves for profile i.

The repeated-outcome SEM is a profile-level path model:

$$A_{ik} = \alpha_k + X_i\beta_k + C_i\gamma_k + \varepsilon_{ik}, \tag{4}$$

where each repeated attacked opinion outcome A_{ik} is regressed on the same profile vector X_i and control vector C_i, while residual covariance between attacked opinion leaves is estimated within the SEM.

The post hoc empirical susceptibility index is not an input variable. It is derived after estimation from the regularized profile aggregation:

$$S_i^* = \mathrm{PctRank}\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{j}\hat{\beta}_{jk}^{(\mathrm{ridge})}X_{ij}\right). \tag{5}$$

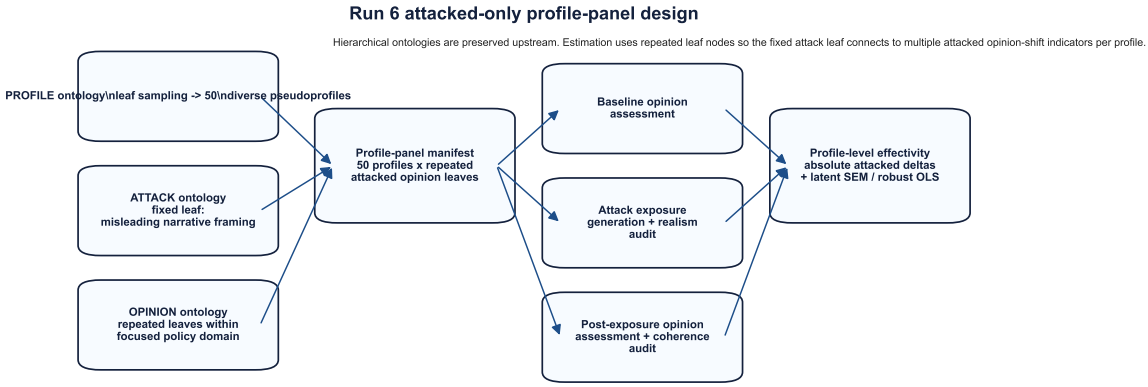
This index is therefore descriptive. It summarizes the model-implied profile propensity toward larger attack effectivity after the run has been completed while shrinking unstable coefficient combinations.

3 Results

Run 6 generated 50 pseudoprofiles and 200 attacked opinion rows. The fixed attack leaf was linked to 4 repeated opinion indicators per profile. Mean absolute attacked shift was 41.65 and mean signed shift was 22.05. Mean attack realism was 0.69, while mean post-exposure plausibility was 0.75.

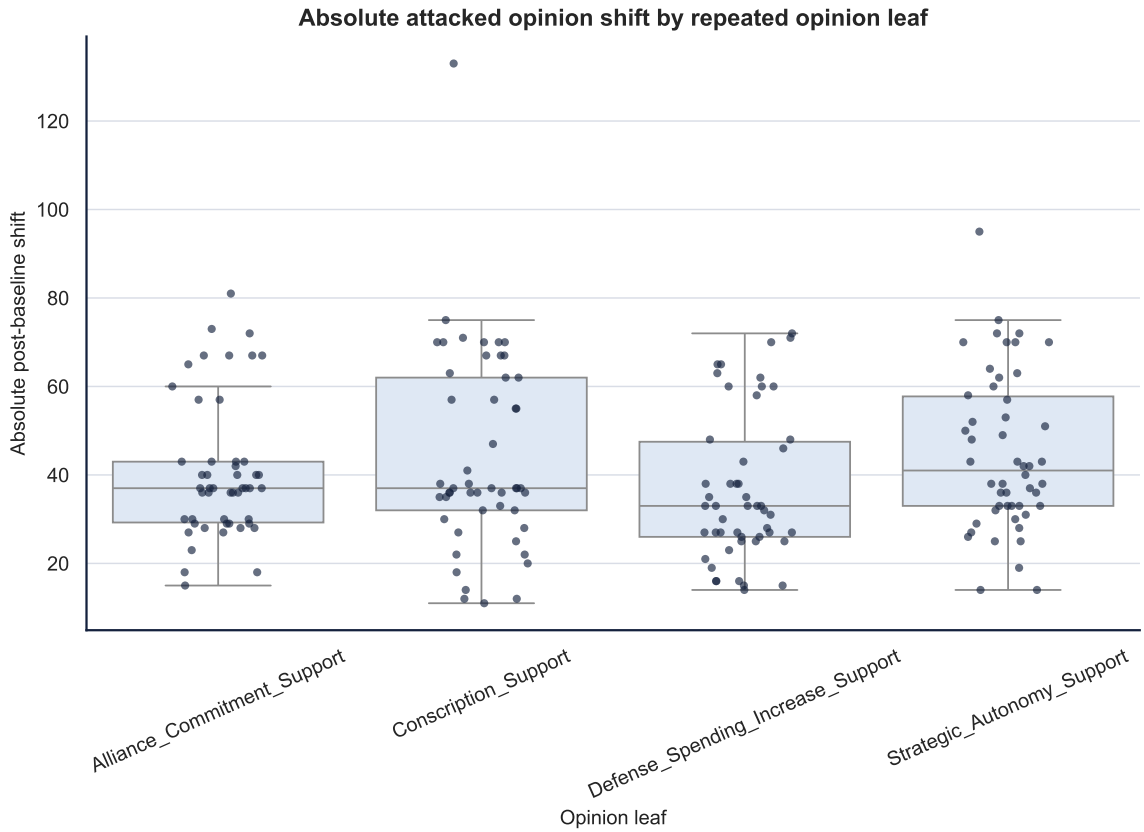
The measurement-and-moderation structure is summarized in Figure 1. Figure 2 shows the distribution of absolute attacked deltas across repeated opinion leaves. Figure 3 shows the descriptive susceptibility weights that summarize the post hoc profile index, and Figure 4 shows the repeated-outcome path-SEM coefficient matrix. The strongest normalized moderator weights in run 6 were Age Years (20.6% of normalized moderator weight); Sex Female (18.0% of normalized moderator weight); Big Five Neuroticism Mean % (14.9% of normalized moderator weight); Sex Other (10.6% of normalized moderator weight). At the ontology-group level, the largest aggregated weight shares were Demographics: Sex (28.6%); Demographics: Age (20.6%); Personality: Neuroticism (14.9%).

Figure 1: Run 6 attacked-only profile-panel design.



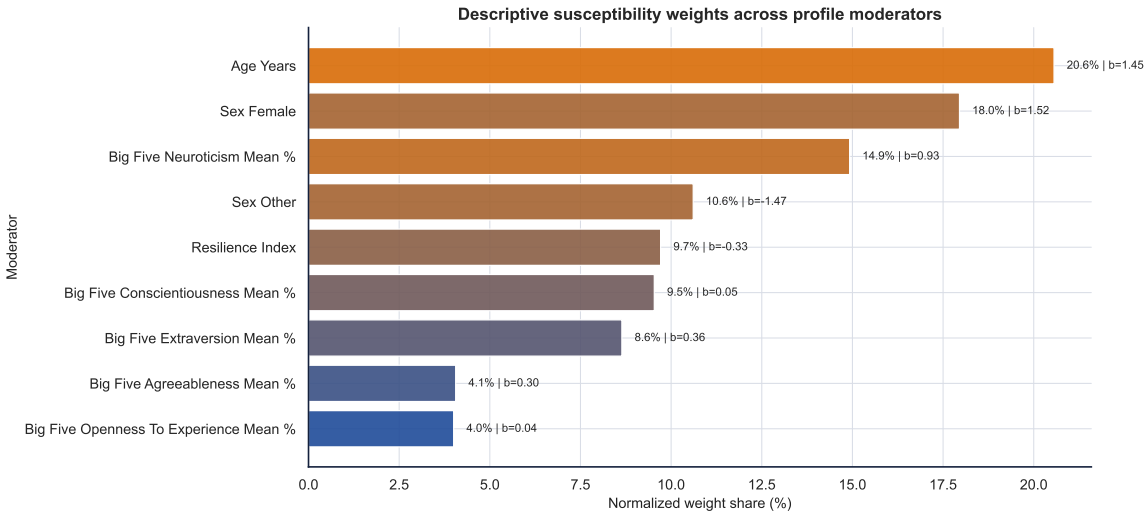
Note. One fixed ATTACK leaf is connected to multiple attacked OPINION leaves for each PROFILE. Hierarchical ontologies are preserved upstream, while estimation is leaf-based and repeated within profile.

Figure 2: Absolute attacked opinion shift by repeated opinion leaf.



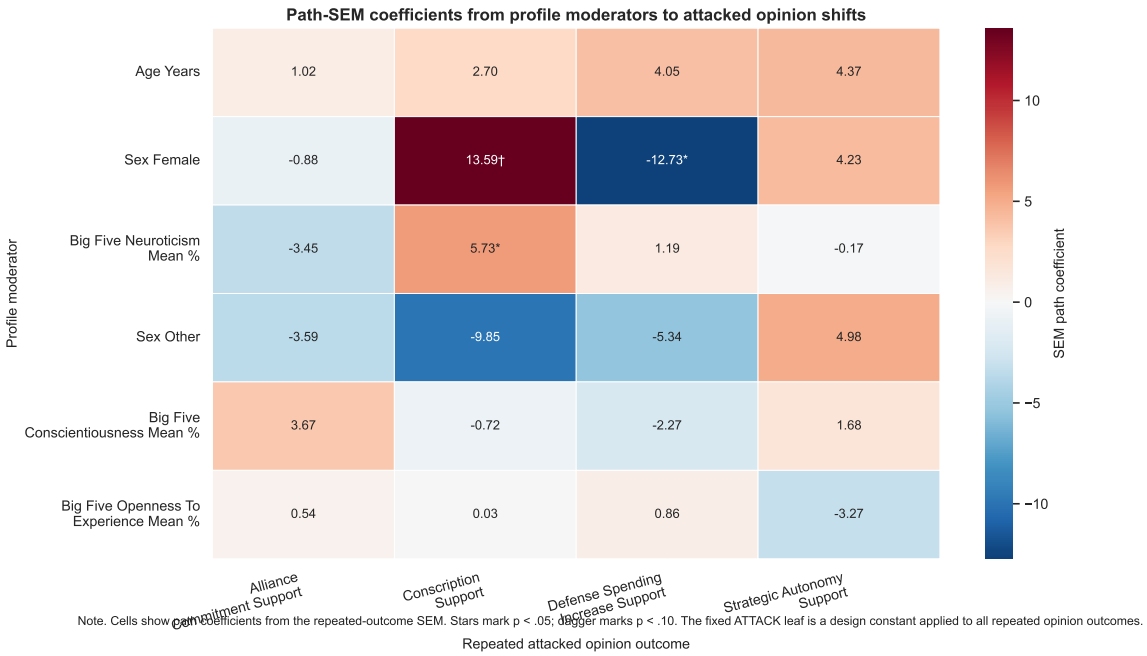
Note. Absolute shift is the primary effectivity outcome because a single attack leaf can generate signed movement in different directions across different opinion leaves.

Figure 3: Descriptive susceptibility weights across profile moderators.



Note. Bars show the normalized contribution of each profile moderator to the post hoc susceptibility index after ridge regularization across the repeated attacked opinion outcomes.

Figure 4: Repeated-outcome path-SEM coefficients across attacked opinion leaves.



Note. Cells show SEM path coefficients from profile moderators to attacked opinion-shift indicators. The fixed ATTACK leaf is constant by design and therefore represented in the study design rather than as a varying regressor.

The repeated-outcome path SEM converged with CFI = 1.000 and RMSEA = 0.000. Those fit indices should still be read cautiously because the model is relatively dense for a pilot of this size and several paths are being screened simultaneously. The clearest leaf-specific paths were Sex Female -> Defense Spending Increase Support (b = -12.73, p = 0.018); Big Five Neuroticism Mean % -> Conscription Support (b = 5.73, p = 0.031); Sex Female -> Conscription Support (b = 13.59, p = 0.053); Big Five Neuroticism Mean % -> Alliance Commitment Support (b =

Table 1: Pilot run design and configuration for the attacked-only profile-panel study.

Parameter	Value
Run ID	run_6
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
Profiles	50
Attacked rows	200
Attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
Opinion domain	Defense_and_National_Security
Repeated opinion leaves	4
Model	mistralai/mistral-small-3.2-24b-instruct

Note. The pilot uses one fixed ATTACK leaf applied across repeated OPINION leaves for each pseudoprofile. All rows are attacked; there is no no-attack control condition in run_6.

Table 2: Attacked effectivity descriptive statistics by repeated opinion leaf.

opinion_leaf_label	n_rows	mean_baseline	mean_post	mean_signed_delta	mean_abs_delta	sd_abs_delta
Strategic_Autonomy_Support	50	434.740	463.980	29.240	44.760	17.614
Conscription_Support	50	-198.540	-168.620	29.920	44.280	22.276
Alliance_Commitment_Support	50	465.380	460.660	-4.720	40.680	15.498
Defense_Spending_Increase_Support	50	267.460	301.200	33.740	36.860	16.660

Note. Absolute shift is the primary effectivity metric for run_6 because the same fixed attack leaf is linked to multiple opinion deltas that can move in different signed directions.

Table 3: Profile moderators of attacked effectivity: controlled contrasts, SEM paths, and descriptive susceptibility weights.

Moderator	Role	Controlled mean b	Controlled p	Ridge mean b	Weight %	SEM mean b	SEM mean |b|	SEM min p	SEM paths
Sex Other	core	-2.918	0.263	-1.465	10.610	-3.450	5.942	0.190	4.000
Sex Female	core	3.246	0.188	1.521	17.954	1.052	7.857	0.018	4.000
Age Years	core	2.592	0.073	1.454	20.561	3.036	3.036	0.132	4.000
Big Five Conscientiousness Mean %	core	0.396	0.755	0.053	9.536	0.590	2.086	0.108	4.000
Big Five Neuroticism Mean %	core	1.039	0.382	0.931	14.924	0.825	2.635	0.031	4.000
Big Five Openness To Experience Mean %	core	-0.916	0.418	0.042	4.004	-0.458	1.174	0.168	4.000
Big Five Agreeableness Mean %	exploratory	1.369	0.211	0.300	4.059				
Big Five Extraversion Mean %	exploratory	-1.151	0.425	0.357	8.642				
Resilience Index	exploratory	-0.419	0.757	-0.330	9.711				

Note. Controlled coefficients summarize moderator associations with mean attacked effectivity. SEM columns summarize the repeated-outcome path model across the attacked opinion leaves. Weight percentages come from the regularized profile aggregation used to compute the post hoc susceptibility index.

-3.45, p = 0.105); Big Five Conscientiousness Mean % -> Alliance Commitment Support (b = 3.67, p = 0.108); Age Years -> Defense Spending Increase Support (b = 4.05, p = 0.132). These p-values are unadjusted and should be interpreted as exploratory. The post hoc susceptibility ranking identified the highest-scoring profiles as profile_0012 (100.0th percentile; mean $|\Delta|$ = 54.2); profile_0105 (98.0th percentile; mean $|\Delta|$ = 46.8); profile_0215 (96.0th percentile; mean $|\Delta|$ = 54.5). The Spearman alignment between the post hoc susceptibility ranking and the observed effectivity percentile was 0.412, which is useful descriptively but should not be over-interpreted as external validation. Controlled summary contrasts and ridge-derived weights together suggested that the most influential profile components in this pilot were Age Years (controlled b = 2.59, p = 0.073, weight = 20.6%); Sex Female (controlled b = 3.25, p = 0.188, weight = 18.0%); Big Five Neuroticism Mean % (controlled b = 1.04, p = 0.382, weight = 14.9%); Sex Other (controlled b = -2.92, p = 0.263, weight = 10.6%); Resilience Index (controlled b = -0.42, p = 0.757, weight = 9.7%).

4 Discussion

Run 6 improves the methodological coherence of the study in three ways. First, the design now directly answers the moderation question under attack rather than drifting into a treatment-versus-control estimand. Second, effectivity is treated as a repeated construct: the same attack leaf is linked to multiple attacked opinion outcomes per profile, and those outcomes remain visible in the SEM instead of being collapsed prematurely. Third, the personal susceptibility index is no longer injected a priori as if it were observed truth. Instead, it is computed post hoc from a regularized aggregation of fitted moderator weights and can therefore be interpreted as a model-derived profile summary.

These changes matter because the research question is not simply whether misinformation can move opinions. The more specific question is which inter-individual differences are associated with larger movement once exposure occurs. In that sense, the attacked-only repeated-leaf design is a cleaner pilot architecture for studying heterogeneity of cyber-manipulation effectivity. The fixed attack leaf reduces attack-side heterogeneity, the repeated opinion leaves reduce item idiosyncrasy, and the moderator-weight table makes the final susceptibility ranking decomposable rather than opaque.

The pilot still has clear limits. The agents are synthetic. The exposure remains one fixed attack leaf. The sample, while larger than earlier pilots, is still modest for high-confidence inference. The path-SEM fit is encouraging but should not be mistaken for definitive construct validation in a near-saturated pilot setting. The post hoc susceptibility index is descriptive because it is derived from fitted model weights rather than observed externally. The current results therefore validate the research workflow more than they establish external truth about real populations.

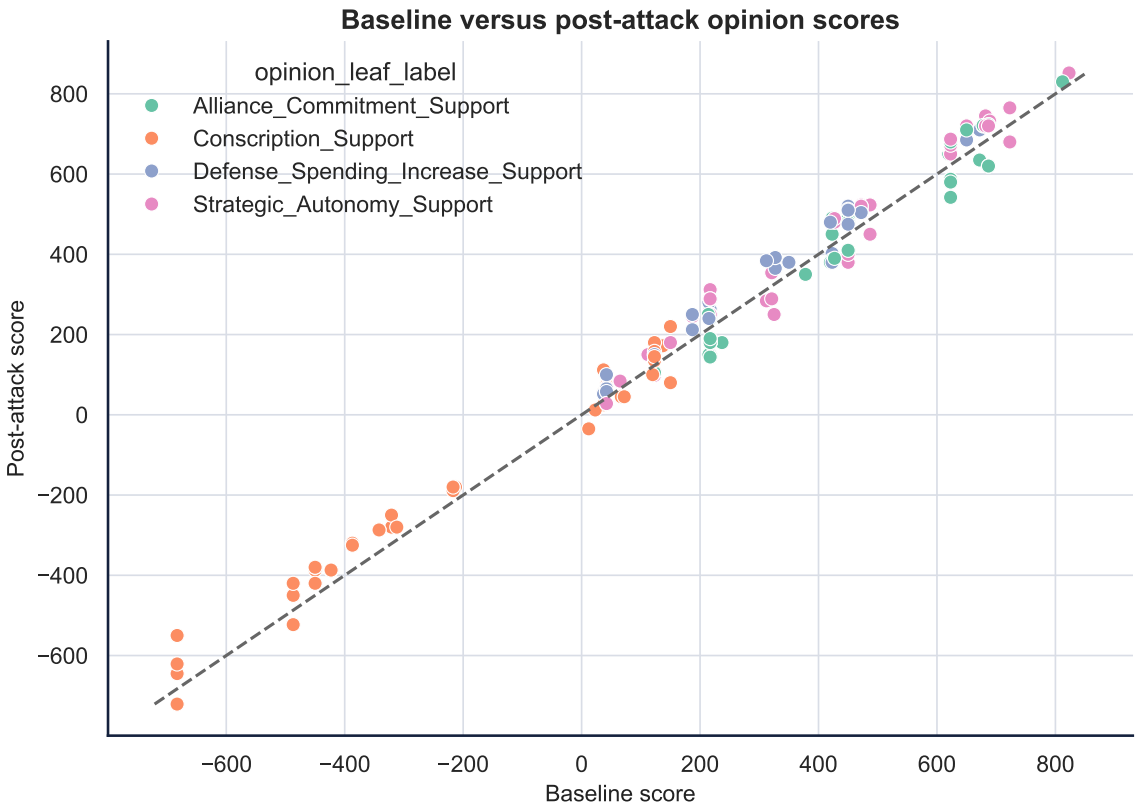
5 Conclusion

This pilot demonstrates that an ontology-driven multi-agent simulation can be structured to study heterogeneity in cyber-manipulation effectivity with substantially better methodological alignment than the earlier pilots. The core move is to treat one fixed attack leaf as producing multiple attacked opinion-shift indicators per profile, estimate those repeated outcomes jointly, and derive the final susceptibility index only after the full profile moderation structure has been

estimated. Run 6 is therefore a stronger methodological answer to the stated research question, even though it remains exploratory and should be scaled further before substantive claims are made.

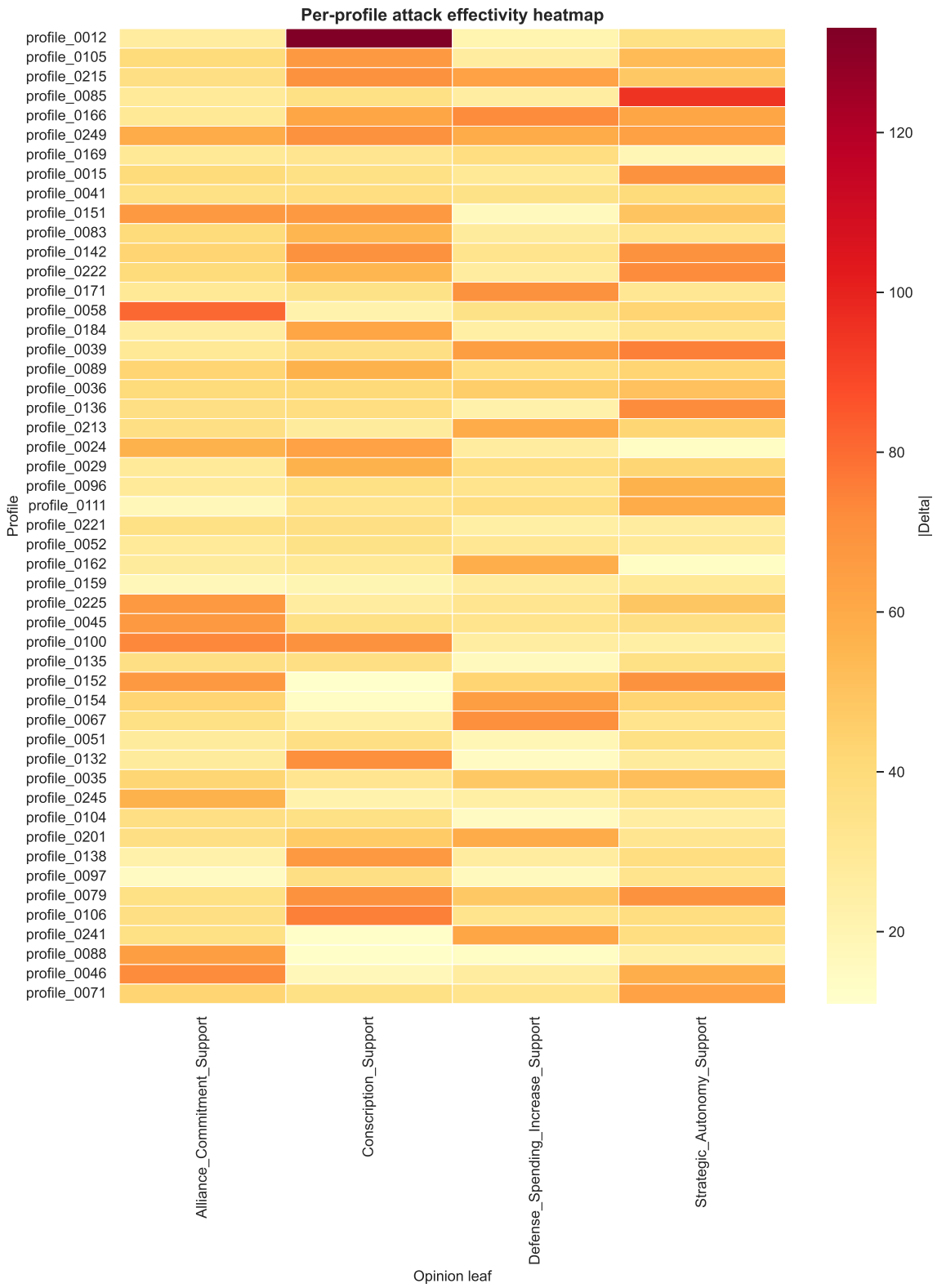
A Supplementary Materials

Figure 5: Baseline versus post-attack opinion scores.



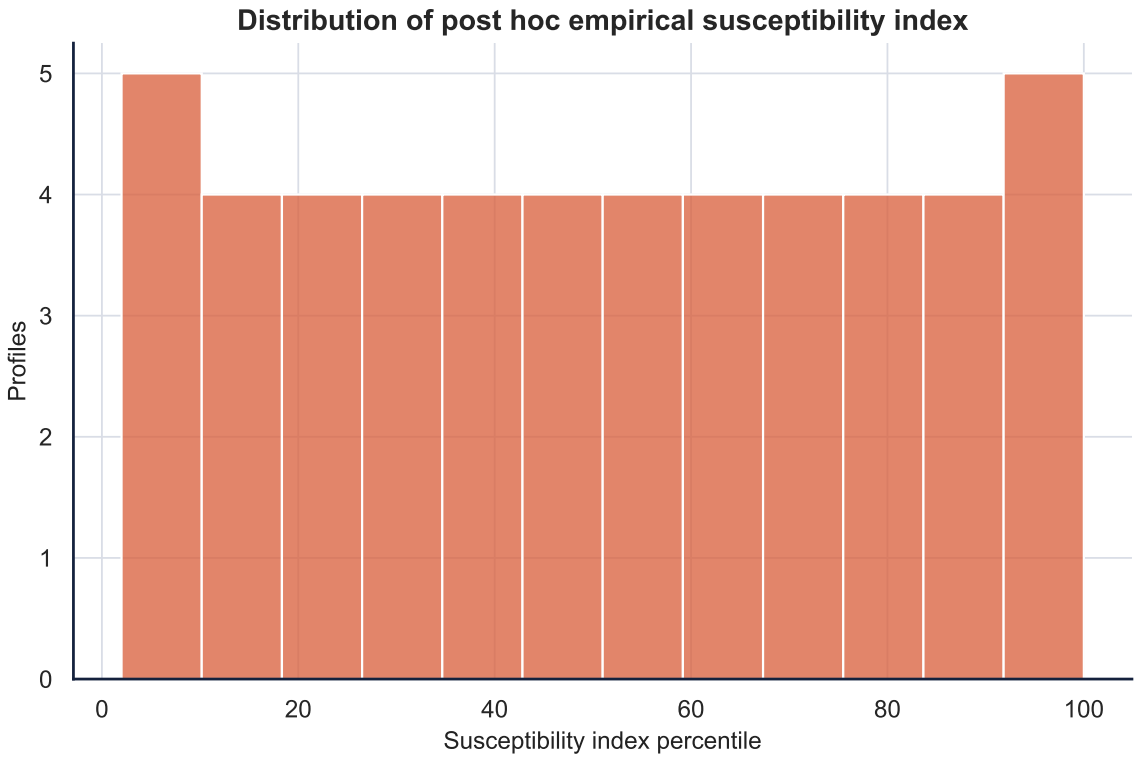
Note. The diagonal marks no change. Points are colored by opinion leaf so issue-specific movement remains visible without collapsing the political state space to a single item.

Figure 6: Per-profile attack effectivity heatmap.



Note. Profiles are ordered by the post hoc empirical susceptibility index. Cells show mean absolute attacked shift for each repeated opinion leaf.

Figure 7: Distribution of the post hoc empirical susceptibility index.



Note. The susceptibility index is the percentile-ranked profile-only linear predictor from the ridge-regularized aggregation of repeated attacked opinion outcomes. It is descriptive rather than independent evidence.

Table 4: Ontology leaves used in the run_6 pilot.

Ontology	Role	Leaf
ATTACK	Fixed pilot leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Alliance_Commitment_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Conscription_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Defense_Spending_Increase_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Strategic_Autonomy_Support

Note. Only ontology leaves are sampled for estimation. The ATTACK ontology contributes one fixed pilot leaf; the OPINION ontology contributes repeated indicator leaves.

Table 5: Core and exploratory profile moderator comparison.

Moderator	Role	Multivariate b	Multivariate p	Univariate b	Univariate p	Ontology group	Normalized weight %
Sex Other	core	-3.040	0.428	-2.918	0.263	Demographics: Sex	10.610
Sex Female	core	1.433	0.707	3.246	0.188	Demographics: Sex	17.954
Age Years	core	376.687	0.876	2.592	0.073	Demographics: Age	20.561
Big Five Conscientiousness Mean %	core	618.949	0.876	0.396	0.755	Personality: Conscientiousness	9.536
Big Five Neuroticism Mean %	core	-1093.951	0.877	1.039	0.382	Personality: Neuroticism	14.924
Big Five Openness To Experience Mean %	core	248.781	0.877	-0.916	0.418	Personality: Openness	4.004
Big Five Agreeableness Mean %	exploratory	0.948	0.486	1.369	0.211	Personality: Agreeableness	4.059
Big Five Extraversion Mean %	exploratory	-0.641	0.708	-1.151	0.425	Personality: Extraversion	8.642
Resilience Index	exploratory	-1309.656	0.876	-0.419	0.757	Composite: Resilience	9.711

Note. Core terms are entered into the latent SEM or primary multivariate profile model. Normalized weight percentages summarize each moderator's share of the total fitted importance after accounting for the moderator's observed variability.

Table 6: Assumption register and peer-review risk register.

Type	Item	Sta
Assumption	Pilot sample size is sufficient for stable moderation estimates.	ok
Assumption	Attack-only design is intentional and aligned with the moderation question.	ok
Assumption	Opinion measurement has enough numeric resolution.	risk
Assumption	Attack text realism is adequate.	risk
Assumption	SEM converges with acceptable fit.	ok
Assumption	LLM outputs are reproducible and auditable.	ok
Critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	add
Critique	Adversarial content may be unrealistic or too generic.	add
Critique	Model dependence on a single LLM could bias outputs.	add
Critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	add
Critique	Attack-only designs cannot estimate the incremental effect of exposure relative to a no-attack counterfactual.	add

Note. The pilot is methodological and exploratory. Risks are surfaced explicitly so later scale-up runs can target the current design bottlenecks.

Table 7: Reproducibility manifest for run_6.

Field	Value
run_id	run_6
paper_title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
output_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/evaluation/r
report_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/research_rep
report_assets_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/research_rep
n_scenarios	200
n_profiles	50
seed	42
attack_ratio	1.000
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
focus_opinion_domain	Defense_and_National_Security
max_opinion_leaves	4
profile_candidate_multiplier	5
primary_moderator	posthoc_profile_susceptibility_index
bootstrap_samples	800
use_test_ontology	True
ontology_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/src/backend
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
temperature	0.150
max_repair_iter	2
profile_generation_mode	deterministic
self_supervise_attack_realism	True
realism_threshold	0.760
self_supervise_opinion_coherence	True
coherence_threshold	0.760
generate_visuals	True
export_static_figures	True
build_report	True
resume_from_stage	08
stop_after_stage	09
save_raw_llm	True
timeout_sec	90
max_concurrency	50
run_stage_checks	False
stage_check_scenarios	3

Note. The manifest captures the full pipeline configuration used to generate the pilot outputs and manuscript assets.

Table 8: Leaf-specific path-SEM coefficients.

Outcome leaf	Moderator	SEM b	SE	p
Alliance Commitment Support	Age Years	1.020	2.819	0.718
Alliance Commitment Support	Big Five Neuroticism Mean %	-3.448	2.129	0.105
Alliance Commitment Support	Big Five Openness To Experience Mean %	0.542	2.272	0.812
Alliance Commitment Support	Big Five Conscientiousness Mean %	3.670	2.286	0.108
Alliance Commitment Support	Sex Female	-0.882	5.638	0.876
Alliance Commitment Support	Sex Other	-3.587	6.030	0.552
Alliance Commitment Support	Mean Baseline Abs Score	-2.291	2.813	0.415
Alliance Commitment Support	Mean Exposure Quality Score	-1.703	2.687	0.526
Conscription Support	Age Years	2.699	3.513	0.442
Conscription Support	Big Five Neuroticism Mean %	5.730	2.654	0.031
Conscription Support	Big Five Openness To Experience Mean %	0.030	2.831	0.991
Conscription Support	Big Five Conscientiousness Mean %	-0.719	2.849	0.801
Conscription Support	Sex Female	13.586	7.026	0.053
Conscription Support	Sex Other	-9.853	7.516	0.190
Conscription Support	Mean Baseline Abs Score	3.545	3.506	0.312
Conscription Support	Mean Exposure Quality Score	-5.744	3.349	0.086
Defense Spending Increase Support	Age Years	4.052	2.690	0.132
Defense Spending Increase Support	Big Five Neuroticism Mean %	1.190	2.032	0.558
Defense Spending Increase Support	Big Five Openness To Experience Mean %	0.859	2.168	0.692
Defense Spending Increase Support	Big Five Conscientiousness Mean %	-2.275	2.182	0.297
Defense Spending Increase Support	Sex Female	-12.728	5.381	0.018
Defense Spending Increase Support	Sex Other	-5.343	5.755	0.353
Defense Spending Increase Support	Mean Baseline Abs Score	3.221	2.685	0.230
Defense Spending Increase Support	Mean Exposure Quality Score	1.219	2.565	0.634
Strategic Autonomy Support	Age Years	4.373	2.941	0.137
Strategic Autonomy Support	Big Five Neuroticism Mean %	-0.174	2.221	0.938
Strategic Autonomy Support	Big Five Openness To Experience Mean %	-3.265	2.370	0.168
Strategic Autonomy Support	Big Five Conscientiousness Mean %	1.683	2.385	0.480
Strategic Autonomy Support	Sex Female	4.233	5.882	0.472
Strategic Autonomy Support	Sex Other	4.984	6.291	0.428
Strategic Autonomy Support	Mean Baseline Abs Score	5.622	2.935	0.055
Strategic Autonomy Support	Mean Exposure Quality Score	-3.846	2.804	0.170

Note. Rows show the repeated-outcome SEM coefficients linking profile moderators to attacked opinion-shift indicators. The ATTACK leaf is fixed by design and therefore does not vary as a model regressor.

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